



Unit 10: REVIEWING and ETHICS

Scientific Research



I. REVIEWING PAPERS

AUTHORS, EDITORS, and REFEREES

- When an **author** completes a paper, it is submitted to the editor of a journal (or the program chair of a conference) for publication.
- The **editor** sends the paper to **referees**, who evaluate the paper and return assessments.
- The editor then uses these assessments to decide whether the paper should be accepted, or, in the case of a journal paper, whether further reviewing or revision is required.



- Authors are expected to be honest, ethical, and careful in their preparation of papers.
- It is ultimately the responsibility of the author—not of the journal, the editor, or the referees—to ensure that the contents of a paper are correct.
- It is also the author's responsibility to ensure that the presentation is at an appropriate standard and that it is their own work unless otherwise stated.

REFEREES



- Referees should be fair and objective, maintain confidentiality, and avoid conflict of interest.
- In addition, they should complete reviews promptly, declare their limitations as referees, take proper care in evaluating the paper, and only recommend acceptance when they are confident that the paper is of adequate standard.
- It would be negligent of a referee to assume that a paper is correct and interesting for superficial reasons such as good writing, impressive mathematics, or author prestige.
- Referees must also ensure that their reviews are accurate and of an appropriate standard.



- The editor's responsibilities are to:
 - ❖ choose referees appropriately,
 - ❖ ensure that the reviewing is completed promptly and to an adequate standard,
 - ❖ arbitrate when the referees' evaluations differ or when the authors argue that a referee's evaluation is incorrect,
 - ❖ and use the reviews to decide whether the paper should be accepted.

CONTRIBUTION

- Contribution is the main criterion for judging a paper.
- A paper is a contribution if it has two properties: *originality* and *validity*.
- The **originality** of a paper is the degree to which the ideas presented are significant, new, and interesting.
 - Most papers are to some degree extensions or variations of previously published work; really groundbreaking ideas are rare.
 - Nonetheless, interesting or important ideas are more valuable than trivial increments to existing work.
 - Deciding whether there is sufficient originality to warrant publication is the main task of the referee.



- The **validity** of a paper is the degree to which the ideas have been shown to be sound.
 - A paper that does no more than claim from intuition that the proposal should hold is not valid.
 - Good science requires a demonstration of correctness, in a form that allows verification by other scientists.
 - Such a demonstration is usually by proof or analysis, modelling, simulation, or experiment, or preferably several of these methods together, and is likely to involve some kind of comparison to existing ideas.

CONTRIBUTION



- When evaluating the significance of a contribution, it is helpful to consider its effect, or **impact**: that is, to judge how much change would follow from the paper being published and widely read.
- If the only likely effect is passing interest from a few specialists in the area, the paper is minor.
- If, on the other hand, the likely effect is a widespread change of practice or a flow of interesting new results from other researchers, the paper is indeed groundbreaking.

EVALUATIONS OF PAPERS

- The process of evaluating a paper involves asking **critical questions** such as:
 - ❖ Is there a contribution? Is it significant?
 - ❖ Are the results correct?
 - ❖ Is the appropriate literature discussed?
 - ❖ Does the methodology actually answer the initial question?
 - ❖ Are the proposals and results critically analyzed?
 - ❖ Are appropriate conclusions drawn from the results, or are there other possible interpretations?
 - ❖ Are all the technical details correct? Are they sensible?
 - ❖ Could the results be verified?
 - ❖ Are there any serious ambiguities or inconsistencies?

EVALUATIONS OF PAPERS

- In addition, there are **further questions** that should be asked of a paper that is under review, which should **not just be correct but should be suitable for the likely audience**.
 - ❖ Is the contribution timely or only of historical interest?
 - ❖ Is the topic relevant to the venue's typical readership?
 - ❖ What is missing? What would complete the presentation? Is any of the material unnecessary?
 - ❖ How broad is the likely readership?
 - ❖ Can the paper be understood? Is it clearly written? Is the presentation at an adequate standard?
 - ❖ Does the content justify the length?

EVALUATIONS OF PAPERS



- Most papers have an explicit or implicit hypothesis—some assertion that is claimed to be true—and a proposal or innovation. **Try to identify what the hypothesis is:** if you can't identify it, there is probably something wrong, and if you can, it will help you to recognize whether all of the paper is pertinent to the hypothesis, and whether important material is missing.
- Referees should make an effort to search for errors that don't affect the quality of the work but should be corrected before going into print.
 - These include spelling and syntax, written expression, errors in the bibliography, whether all concepts and terms have been defined or explained, errors in any formulas or mathematics, and inconsistency in just about anything from variable names to table layout to formatting of the bibliography.
 - In particular, only a referee is likely to find errors in mathematics.

EVALUATIONS OF PAPERS



- Real incompetence in the **presentation** is grounds for rejection, because a paper is of no value if it cannot be read.
- However, excellence in presentation does not justify acceptance.
 - ❖ Occasionally a referee receives a paper that is well written and shows real care in the development of the results, but which does no more than reproduce existing work. Such papers must be rejected.
- As a consequence of the peer review system, active researchers should expect to referee about two to three times as many papers as they submit (or somewhat less if their papers are usually co-authored) and only decline to referee a paper with good reason.

CONTENT OF REVIEWS

- Reviewing of papers serves two purposes.
 - ❖ The explicit purpose is that it is the mechanism used by editors to decide whether papers should be accepted for publication.
 - ❖ The implicit purpose—equally important, and often overlooked—is that it is a means of sharing expertise between scientists, via comments for the authors.
- Reviews usually include other things besides written comments (such as scores on certain criteria, which are used to determine whether the paper should be accepted), but it is the comments that authors find valuable.
- The review should make some kind of case about the paper: whether it is of an adequate standard and what its flaws are. That is, it is an analysis of the paper, explaining why it is or is not suitable for publication.



II. ETHICS

- Science is built on **trust**.
 - Researchers are expected to be honest, and research is assumed to have been undertaken ethically.
 - For example, referees assess whether results are significant but are not expected to investigate whether the reported experiments actually took place, because it is assumed that the authors have not lied about their work.



- In brief, the scientific community expects:
 - published research to be new, objective, and fair;
 - researchers should not present opinion as fact, distort truths, or imply that previously published results are original;
 - research should be undertaken within relevant ethical frameworks, which protect privacy and minimize the risk of harm to individuals;
 - and researchers should not plagiarize others or misrepresent their contributions to the work.
- However, there is no international academic police force, rule of law, or investigative agency. Ethical issues are largely managed by individual adherence to the expectations of academic community, and through the actions of self organizing groups that respond to ethical breaches when they do occur.

FORMS OF MISCONDUCT

- The most conspicuous form of unethical behavior in computing is **plagiarism**, because it steals work from other scientists and the hurt to others is obvious.
- One is **abuse of power**, such as when senior academics insist on being listed as authors of papers they have not contributed to.
- Another is **fraud**, in which claimed results were not in fact observed, or are much less substantial than was claimed.
- Issues of ethical concern for science in general include misrepresentation, plagiarism and self-plagiarism, authorship, confidentiality and conflict of interest, harassment and abuse of power, and use of human subjects.

INTELLECTUAL CREATIONS

- **Intellectual creations:** the stuff that knowledge workers make, including concepts, inventions, discoveries, designs, documents (text, images, or video), or code.
- People might own their intellectual creations, or the creations might belong to their employer, or a publisher.
- Some forms of intellectual creation have an established legal framework.
 - Informally, **copyright** concerns the expression of ideas; that is, the words and pictures used to say something are protected, but not the ideas themselves.
 - **Patents** and *prior invention* protect processes, inventions, and concepts; that is, the ideas, but not how they are described.

INTELLECTUAL CREATIONS



- When a researcher writes a paper, the content represents new intellectual property, which may be patentable.
- The copyright in the form or presentation is held by the author; typically, when the paper is published the author assigns copyright to the publisher.
- Thus the publisher owns the text, figures, and diagrams, and the author—or anyone else—cannot legally use these again without the publisher's written permission.
- Specifically, copyright prohibits use without permission of pictures and diagrams, or of substantial volumes of text.
- Brief pieces of text can be used so long as they are attributed, that is, quoted.

1. PLAGIARISM

- **Plagiarism** is re-use in one paper of material that has appeared in another, without appropriate acknowledgement.
- The theft may involve **ideas, illustrations, results, text**, or even whole papers; and includes, not just copying from published papers, but from material in electronic form, such as Web pages, news articles, or email.
- By plagiarizing, a researcher hopes to obtain credit for work that has already been published, and not necessarily by someone else.
- However, while some people do make a deliberate decision to steal and there is a complex range of factors that lead people to plagiarize, one cause seems particularly common: **misjudgement by an inexperienced researcher**.

1. PLAGIARISM

- Such misjudgements can arise when a research student is unaware of appropriate academic style.
- For example, a researcher investigating B-trees may find an elegant illustration in a textbook and decide that it is perfect for a forthcoming paper; but copying this illustration (either by reproducing it or by imitating it) is plagiarism.
- Similarly, a researcher describing B-trees may feel that a paragraph in a reference cannot be improved on; but copying it verbatim is plagiarism. Even a close paraphrase of it is likely to be plagiarism.

1. PLAGIARISM

Another form of misjudgement is **inappropriate or inadequate citation**.

Original Barlman and Trey (2001)	The impact of viruses has become a major issue in many large organizations, but most still rely on individual users maintaining virus definitions, with no internal firewalls to protect one user from another. However, any structure is only as strong as its weakest link; these organizations are highly vulnerable.
Plagiarism ✗	Viruses have become a major issue in many large organizations, but most organizations still rely on users maintaining virus definitions on their individual computers, with no internal firewalls to protect one computer from another. However, any structure is only as strong as its weakest link; these organizations are highly vulnerable to infection (Barlman and Trey 2001).
Appropriate citation ✓	As discussed by Barlman and Trey (2001), who investigated the impact of viruses in large organizations, “most still rely on individual users maintaining virus definitions, with no internal firewalls to protect one user from another. However, any structure is only as strong as its weakest link; these organizations are highly vulnerable.”
Appropriate citation ✓	Barlman and Trey (2001) investigated the impact of viruses in large organizations. They found that organizations are vulnerable if individuals fail to keep virus definitions up to date, as internal firewalls are rare.

1. PLAGIARISM

- **Paraphrase of the structure of a paper is also plagiarism.**
- If one paper follows another to the extent that they:
 - use the same headings,
 - have tables of the same layout,
 - cite much the same background literature,
 - describe the literature with respect to the same criteria,
 - and have similarly designed experiments with similar data exploring the same properties,

then the second paper is arguably plagiarized.

2. SELF-PLAGIARISM

- Authors who re-use their own text may well be plagiarizing.
- Some scientists feel that it is acceptable to re-use their own background material from paper to paper. However, there are both principled and pragmatic arguments against this practice.
- Publication of more than one paper based on the same results is prohibited under the standard scientific codes of conduct.
 - In this context, “the same results” does not necessarily mean a particular experimental run; if an experiment has been tried on some data, running the same experiment on other data is not new work unless it leads to new conclusions.
- Self-plagiarism can also be considered from the point of view of copyright. As noted earlier, when you publish a research paper the copyright is usually assigned to the publisher, who thus owns, not the ideas, but your expression of them.

3. MISREPRESENTATION

- **Misrepresentation** occurs when a paper does not accurately reflect the outcomes that were observed or the contributions of previous research.
- When presenting results, researchers are expected to:
 - ❖ ensure that they are accurate,
 - ❖ describe any experimental issues or limitations that could have affected the outcome,
 - ❖ provide enough detail to enable reproduction or verification,
 - ❖ be fair in description of other work,
 - ❖ report negative as well as positive results,
 - ❖ not state falsehoods,
 - ❖ and take the effort to ensure that statements are complete and accurate.

3. MISREPRESENTATION



- In its clearest form, misrepresentation is fraud: the making of claims that are outright false.
- Another form of misrepresentation is when authors imply that they have high confidence in their results when in fact the experiments were preliminary or were limited in some way.
 - ❖ For example, reported running times may be based on a small number of runs with high variance, or there may be uncertainties about the quality of the implementation.
- Other forms of misrepresentation concern interpretation of past work.
 - ❖ It can be tempting for authors to exaggerate the significance and originality of their results, and to diminish the status of previous results in the field, to increase the likelihood of their work being published.

4. AUTHORSHIP

- A broadly accepted view is that each author must have made some significant contribution to the intellectual content of the paper.
- Thus directed activities such as programming do not usually merit authorship, nor does proofreading.
- An author should have **participated in the conception, execution, or interpretation of the results**, and usually an author should have participated to some degree in all of these activities.
- A researcher who has contributed to the research must be given an opportunity to be included as an author, but authors should not be listed without their permission.

5. CONFLICT OF INTEREST

- Researchers should not referee a paper where there is a possible conflict of interest, or where there is some reasonable likelihood that it will be difficult for the referee to maintain objectivity;
- Examples include papers by:
 - ❖ a recent advisor, student, or co-author of the referee,
 - ❖ or an author with whom the referee recently had close interaction, including not only personal or employment relationships but also situations such as competition for an appointment.
- It can be difficult to maintain objectivity if the author's opinions strongly conflict with your own. Make every effort to be fair, or seek an alternative referee.

6. CONFIDENTIALITY



- A related issue is of confidentiality: papers are submitted in confidence and are not in the public domain.
- Papers you are reviewing should not be shown to colleagues, except as part of the refereeing process; nor should they be used as a basis for your own research.
- In practice there is something of a grey area—it is impossible not to learn from papers you are refereeing, or to ignore the impact of their contents on your own work.
- Nonetheless, the confidentiality of papers should be respected.



III. ETHICAL ISSUES TO CONSIDER CONCERNING RESEARCH PARTICIPANTS

1. COLLECTING INFORMATION



- One could ask:
 - why should a respondent give any information to a researcher?
 - What right does a researcher have to knock at someone's door or to send out a questionnaire?
 - Is it ethical to disturb an individual, even if you ask permission before asking question?
 - Why should a person give you his/her time?
 - Your request for information may create anxiety or put pressure on a respondent. Is this ethical?

1. COLLECTING INFORMATION



- Research is required in order to improve conditions.
- Provided any piece of research is likely to help society directly or indirectly, it is acceptable to ask questions, if you first obtain the respondents' **informed consent**.
- Before you begin collecting information, you must consider the relevance and usefulness of the research you are undertaking and be able to convince others of this also.
- If you cannot justify the relevance of the research you are conducting, you are wasting your respondents' time, which is unethical.

2. SEEKING CONSENT



- In every discipline it is considered unethical to collect information without the knowledge of participants, and their expressed willingness and informed consent.
- Informed consent implies that subjects are made adequately aware of:
 - ❖ the type of information you want from them,
 - ❖ why the information is being sought,
 - ❖ what purpose it will be put to,
 - ❖ how they are expected to participate in the study,
 - ❖ and how it will directly or indirectly affect them.
- It is important that the consent should also be voluntary and without pressure of any kind.

2. SEEKING CONSENT

“Under standards set by the National Commission for the Protection of Human Subjects, all informed-consent procedures must meet three criteria: participants must be competent to give consent; sufficient information must be provided to allow for a reasoned decision; and consent must be voluntary and uncoerced.”

Schinke, Steven P. & Lewayne Gilchrist, 1993, 'Ethics in research', in R. M. Grinnell (eds), *Social Work, Research and Evaluation* (4th edn), Itasca, IL, F.E. Peacock.

3. SEEKING SENSITIVE INFORMATION

- Certain types of information can be regarded as sensitive or confidential by some people and thus an invasion of privacy.
- Asking for this information may upset or embarrass a respondent.
- For most people, questions on sexual behaviour, drug use and shoplifting are intrusive. Even questions on marital status, income and age may be considered to be an invasion of privacy by some
- It is not unethical to ask such questions provided that you clearly and frankly tell your respondents the type of information you are going to ask, and give them sufficient time to decide if they want to share the information with you, without any major inducement.

4. THE POSSIBILITY OF CAUSING HARM TO PARTICIPANTS

Harm includes “not only hazardous medical experiments but also any social research that might involve such things as discomfort, anxiety, harassment, invasion of privacy, or demeaning or dehumanizing procedures.”

Bailey, Kenneth D., 1978, *Methods of Social Research* (3rd edn), New York, Free Press.

4. THE POSSIBILITY OF CAUSING HARM TO PARTICIPANTS

- When you collect data from respondents or involve subjects in an experiment, you need to examine carefully whether their involvement is likely to harm them in any way.
- If it is, you must make sure that the risk is minimal. Minimum risk means that the extent of harm or discomfort in the research is not greater than that ordinarily encountered in daily life.
- It is unethical if the way you seek information creates anxiety or harassment, and if you think it may happen, you need to take steps to prevent this.

5. MAINTAINING CONFIDENTIALITY



- Sharing information about a respondent with others for purposes other than research is unethical.
- You need to make sure that at least the information provided by respondents is kept anonymous.
- It is unethical to identify an individual respondent and the information provided by him/her. Therefore, you need to ensure that after the information has been collected, its source cannot be identified.
- It is unethical to be negligent in not protecting the confidentiality and anonymity of the information gathered from your respondents.

REFERENCES



- Zobel, J. (2014). *Writing for computer science*. London: Springer.
- Kumar, R. (2011). *Research methodology: a step-by-step guide for beginners*. SAGE.